
UNIT 16 MOBILE OPERATING SYSTEMS AND DEVELOPMENT ENVIRONMENTS

- 16.0 Introduction To Mobile Operating Systems
 - 16.0.1 Key Concepts And Functions Of An Operating System
- 16.1 Objectives
- 16.2 Application Programming Interface
- 16.3 Linux For Mobile Devices
- 16.4 Development Process
 - 16.4.1 Requirements Elaboration Phase
 - 16.4.2 Design Phase
 - 16.4.3 Implementation/Develop Phase
 - 16.4.4 Testing Phase
 - 16.4.5 Deployment Phase
- 16.5 Development Tools And Emulators
- 16.6 Apple iOS
 - 16.6.1 iOS Architecture
- 16.7 Android
 - 16.7.1 Android Architecture
- 16.8 Differences Between Ios And Android Operating Systems
- 16.9 Summary
- 16.10 Solutions/Answers
- 16.11 Further Readings

16.0 INTRODUCTION TO MOBILE OPERATING SYSTEMS

The operating system is a system software that manages the interactions of the user with the underlying hardware. The mobile operating system is mainly designed for the mobile devices.

Mobile OS normally provides an integrated development environment, a development tool and an emulator. Some mobile OS platforms such as Apple iOS are specifically designed for underlying device and few others mobile OS platforms such as Android works across multiple mobile devices.

Mobile OS handles multiple user inputs and submit them as tasks for the underlying CPU. Often the mobile OS has to do multitasking when the user simultaneously uses multiple apps. The mobile OS essentially abstracts the hardware details of the underlying mobile device. The mobile OS also enables the end user to interact with various mobile device features such as sensors, cameras , wireless services and so on.

Many mobile OS platforms provide inbuilt apps such as cameras, contacts, messengers and so on. Android and Apple IOS for the two most popular mobile OS in the world.

16.0.1 Key concepts and functions of an operating system

In this section let us look at the key concepts of an operating system.

1. Process and Thread management

Process and thread management are one of the key functions of an operating system.

A process is a software program that runs on the CPU and uses the memory for computation. The OS allocates the CPU resources and memory resources for the process. An executing program is called process. When the process is first created it will be in the “created” state. Later on, a process moves to various States such as activated, running, suspended, deactivated and deleted state. When one process sends a message to another process, it is called “inter process communication”. When the OS allocates the CPU and memory for the process, it is called a task.

The mobile OS manages both system processes and user processes. This includes creation of system processes and user processes; suspension and resumption of processes; process synchronization and deadlock handling.

A process is usually heavy weight as it needs CPU, memory and other resources. Thread is a lightweight process as it needs minimal resources. A process may consist of various threads. We normally launch multiple threads for parallel computation. Similar to a process, even a thread has many states such as creating, blocking, running, deleting, suspending and deleting.

2. Memory management

The operating system manages the system memory by allocating appropriate memory blocks for each of the processes. The memory consists of a large array of words or bytes each with an unique memory address. The mobile OS reads the sequence of words or bytes required for the process. The mobile OS also keeps track of the used memory and unused memory and prioritizes the processes that need to be loaded into the memory.

The operating system also de-allocates the memory blocks when the program has completed its job. Additionally the OS also manages the memory buffers shared memory and handles the efficient paging for the programs.

3. Device management

A device is a physical entity of the mobile. Few examples of devices are Camera, sensor, socket port or a memory buffer. The mobile OS manages the device throughout its usage life cycle. To start with, the mobile OS creates a device ID and allocates distinct memory blocks to the device. the mobile OS then opens the device for operations such as reading and writing. The mobile OS also reads the incoming data on devices such as sockets and ports. The mobile OS reads and writes the data for the device. The mobile OS also closes or de-registers the device when it is not required and deletes the memory allocated to the device.

4. File management

A file essentially stores the data as a record on the devices such as disk or flash memory. Each record has a name, descriptor and a memory block allocated to it. The mobile OS abstracts the internal storage units as files. The mobile OS manages various functions related to the file such as reading, writing, deleting,

updating and others. Optionally the files are also backed up based on the requirements. The files are organized as logical folders for easier management.

The mobile OS manages various functions of the file such as file creation, file opening, file update and file deletion. When the user wants to create a new file the mobile OS will add a file descriptor after creating a new file. The file descriptor consists of various details like the file extension, the file size and file metadata such as the creation time, permissions, author's name and so on. The OS opens the file in read or write mode. During the write mode, the program updates or adds new records.

The mobile OS also manages the security of the files and folders. Using the permission model, the OS ensures only the appropriate users with required permissions can access the files and folders.

5. Input/output management

The mobile OS manages various input/ output devices such as keyboard, printer, USB Port and others. The OS Creates a unique device ID for each of the devices and opens the device for read and write operations. The mobile OS also reads the data from the device and writes the data to the device and manages the buffer.

6. Network management

The mobile OS manages various network components such as the Wi-Fi, LTE , GSM and CDMA connectivity.

The enterprise applications would also interact with other platforms such as database and middleware platforms. The applications can use the JDBC API to interact with the database. The middleware platforms expose standards compliant API such as REST APIs, HTTP APIs, SOAP APIs and websocket APIs.

16.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of mobile operating systems,
- understand the application programming interface,
- understand on the role of Linux for mobile devices,
- understand the end to end development process,
- understand the main development tools and emulators
- understand the key mobile OS platforms such as Android and Apple iOS and the differences between them.

16.2 APPLICATION PROGRAMMING INTERFACE

The mobile OS provides various applications and system functions for the mobile apps. For instance the mobile OS provide functionality such as call, SMS, Email, calendar, contacts and others. The mobile OS abstracts the system details and hardware interfaces through these APIs. For instance, without knowing the full specifications of the mobile camera, the mobile app developer can invoke the camera APIs for taking the pictures.

The mobile OS exposes the Application programming interface (API) for each of the system functions. The mobile app developers can use the APIs that are based on the well-defined contract.

The mobile OS also provides the APIs for creating other features such as user interface, voice based interfaces and such.

16.3 LINUX FOR MOBILE DEVICES

Few mobile OS such as Apple iOS are built for specific Apple hardware devices. However Android works on vast majority of mobile devices. Android uses Linux, an open source OS that can be customized to work on various hardware devices.

Majority of the mobile device manufacturer supports Linux which can use Android. The core features such as Opera web browser, wireless features, camera, networking support are fully supported on Linux.

16.4 DEVELOPMENT PROCESS

The development process mainly consist of requirements elaboration phase, design phase, implementation phase and testing phase and deployment phase.

We have provided all the phases of the development process in Figure 1.

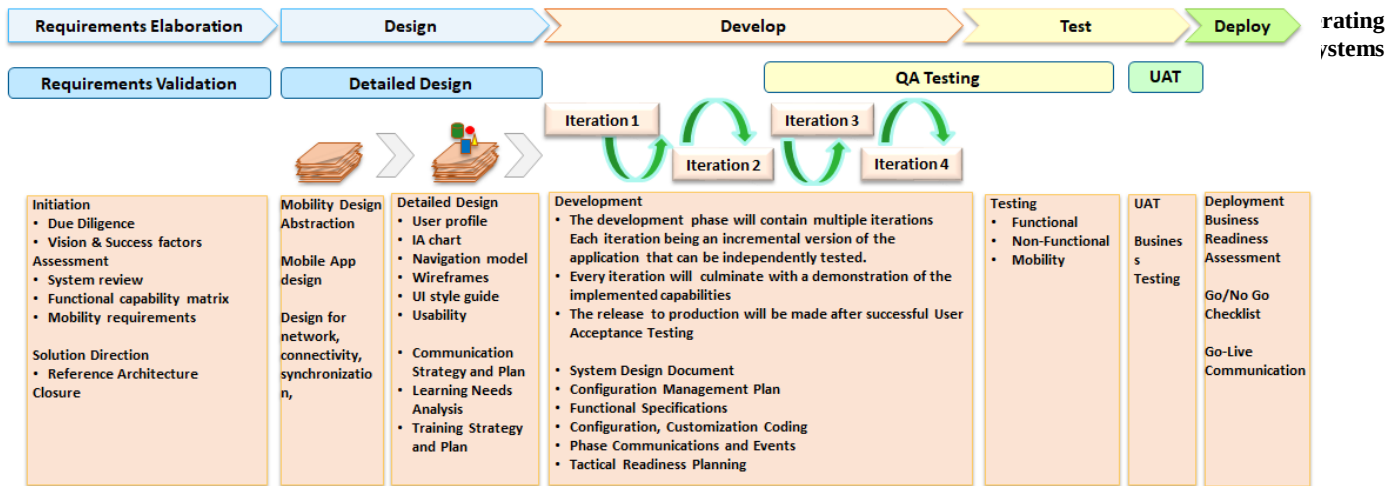


Figure 1 Phases in the development process

We shall look at each of the phases in detail.

16.4.1 Requirements Elaboration phase

We have depicted the key activities in the requirements elaboration phase in Figure 1.

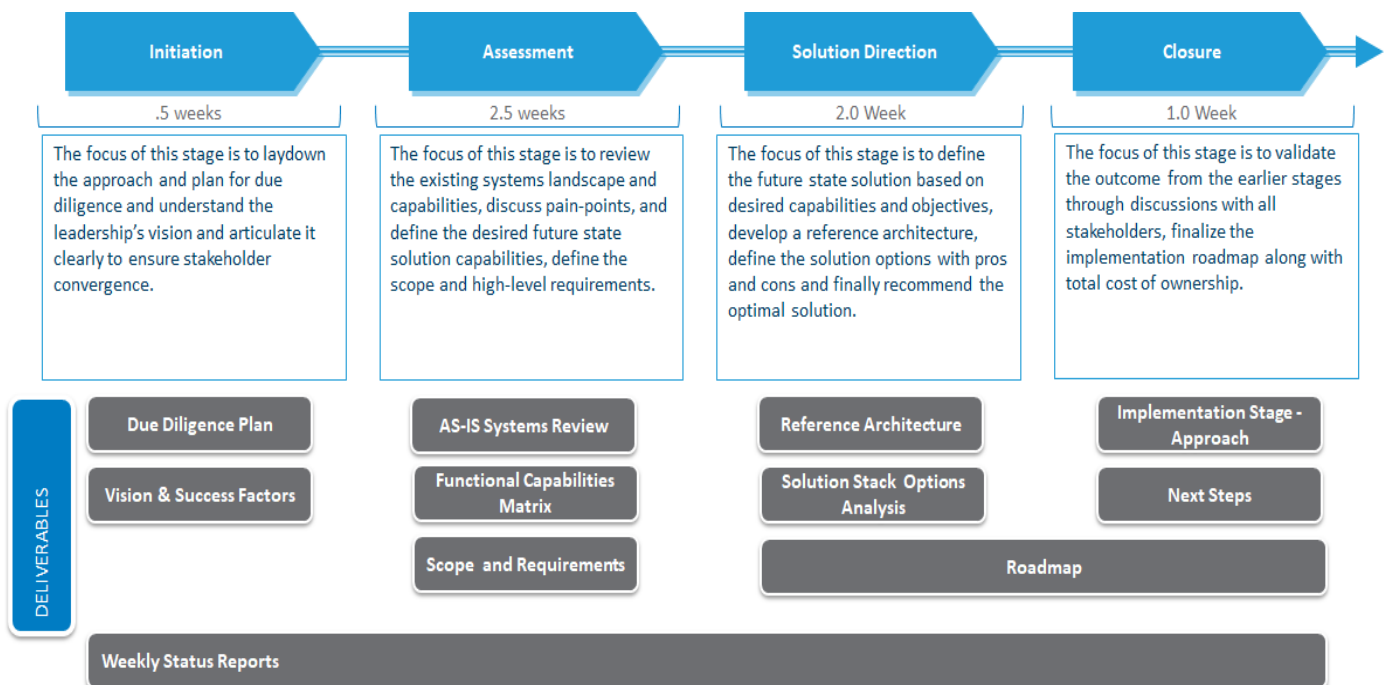


Figure 2 Requirements Elaboration Phase

During the requirements elaboration phase, we have various stages. The first stage is the initiation stage and we have given below are the main activities:

- Discuss and finalize the detail plan for the 6 week assessment
- Meet the program leadership to understand their vision for the future state
- Understand the pain-points from the current solution, discuss/document the key success factors for the future state solution.
- Meetings/Interviews with the business stakeholders to understand the short & long-term objectives of the solution

- Create a list of all internal and external systems for integration, and the key stakeholders to be included as a part of the future state planning meetings

The main deliverable from this stage are as follows:

- Detailed Project Plan
- Vision, and Key Success Factors

The next stage is the assessment stage where we do the assessment of the current technical eco-system. Given below are the key activities in this stage:

- Assess the as-is technical environment to understand the current solution, technologies, customizations, integrations etc. spanning portal, content management, search, social networking and backend applications.
- Meetings/interviews with business and IT stakeholder to understand the business functional capabilities for the future state
- Assess the key mobility related capabilities:
 - The type of mobile devices to be supported
 - The form factor of the mobile devices
 - Understand various constraints such as network/bandwidth constraints
 - Understand the multi-lingual capabilities required for the mobile device
 - Understand the various networks that need to be supported such as GSM, Wi-Fi, CDMA, GPRS etc.
 - Understand various connectivity requirements
- Understand various data sources and user inputs such as multi-touch inputs, gestures, face recognition, biometric authentication and such.
- Understand the data integration requirements such as Email server, SMS server etc.
- Develop the capability matrix for the following functional areas -
 - Mobile capabilities
 - Content Management
 - Search
 - Social networking and collaboration
 - Integration requirements
 - Personalization
 - Tasks & Workflow management
- Develop the high-level scope and requirements for the future state solution
 - Functional grouping and segregation
 - Business value driven initial prioritization on the grouped functionalities

The deliverable of this phase are as follows:

- As-is systems review document
- Functional Capability Matrix, Groupings and Prioritization
- High-Level Scope and Requirements documentation

We evaluate and shortlist various solution options in the solution direction phase. Given below are the key activities in this stage:

- Confirm the relevance of various planned and in-flight programs critical for the success of this program. Example: dependency on external systems

- Develop the reference architecture based on the high-level scope and requirements and determine dependencies.
- Identify & finalize the future state solution stack options for the mobility solution. Evaluate the native mobile app along with hybrid mobile app. Identify all related and dependent products/frameworks required for the solution.
- Define and document the pros & cons for the solution options, and provide recommendations – both for hardware and software.
- Define the total cost of ownership – based on license fee, implementation and support cost.
- Develop the implementation roadmap based on the recommended option, and mark out the dependencies

Given below are the deliverable of this phase:

- Reference Architecture
- Solution Options, and Recommendations (including TCO analysis)

In the last step of closure, we provide the recommendations and given below are the various activities:

- Present summary of the findings so far – validate with stakeholders
- Discuss, and finalize the solution option for the implementation phase
- Complete the implementation roadmap, timelines and dependencies
- Close out *the* stage
- Develop the implementation roadmap based on the recommended option, and mark out the dependencies

Given below are the deliverable from this stage:

- Implementation Roadmap
- Final set of deliverables as detailed in the prior stages

16.4.2 Design phase

During design phase we do the solution design. The key design principles are as follows:

1. Creating decoupled components
2. Creating modular components
3. Designing components that have lower coupling
4. Creating components based on single responsibility principle.

The main activities in the design phases are given below

User Research

- Stakeholder interviews to understand business goals, user profiles, needs, and pain points
- Working sessions with actual users as identified by business stakeholders
- Prioritization of user scenarios & user profiles

Conceptual Design

- Information architecture
- Navigation flows
- Task flows
- High-level wire-frame for scenario

Detail Design

- Interaction and Interface design for identified scenarios in wire-frames
- Iterative design with feedback from business team

Visual Design

- Depiction of end-state visual design of unique page types with brand guidelines.
- We also design the screens for various mobile device form factors

Mobile app core design

This includes the design for various connectivity types and for data synchronization.

Integration Design

We design the most appropriate integration methodology for various data sources such as enterprise applications, web services, databases and such.

We also design the strategy for caching, data synchronization and such

The main deliverable from this phase are as follows:

- User profile documentation
- IA chart
- Navigation model
- Wireframes for selected user scenarios
- Static JPG images of visual design for unique page types
- UI style guide document

The detailed steps in the visual design is depicted in Figure 3

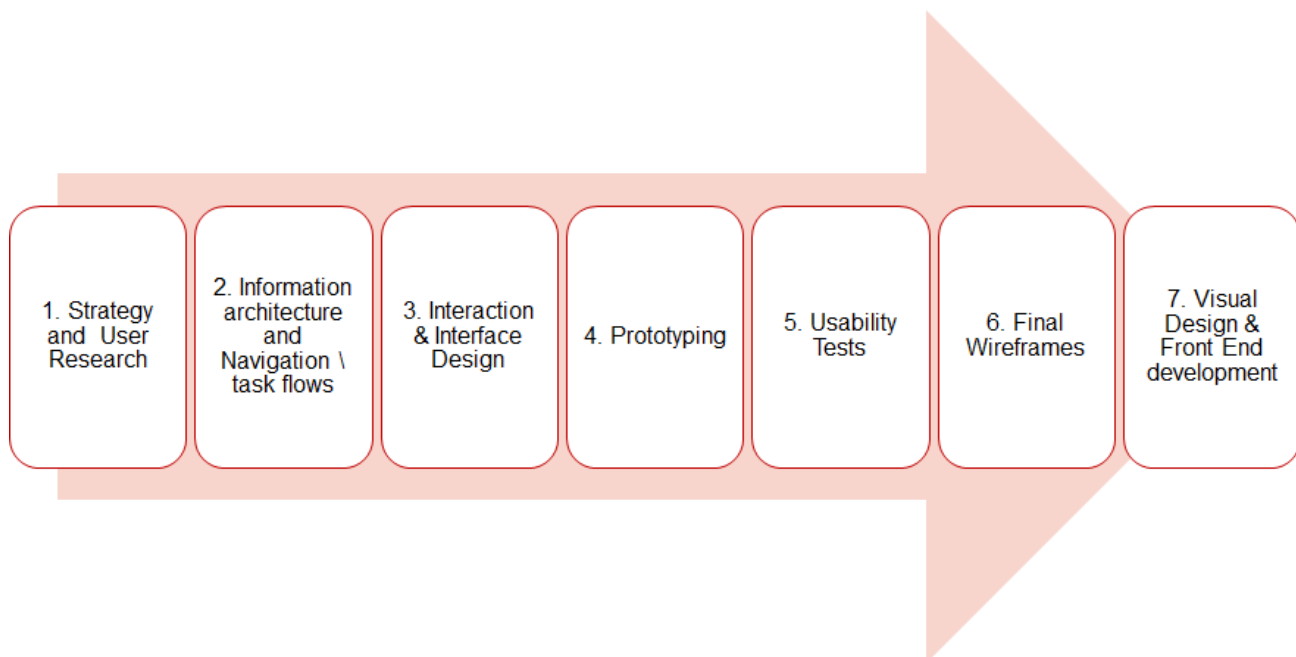


Figure 3 Visual Design Steps

The main deliverable of the user interface design is depicted in figure 4.

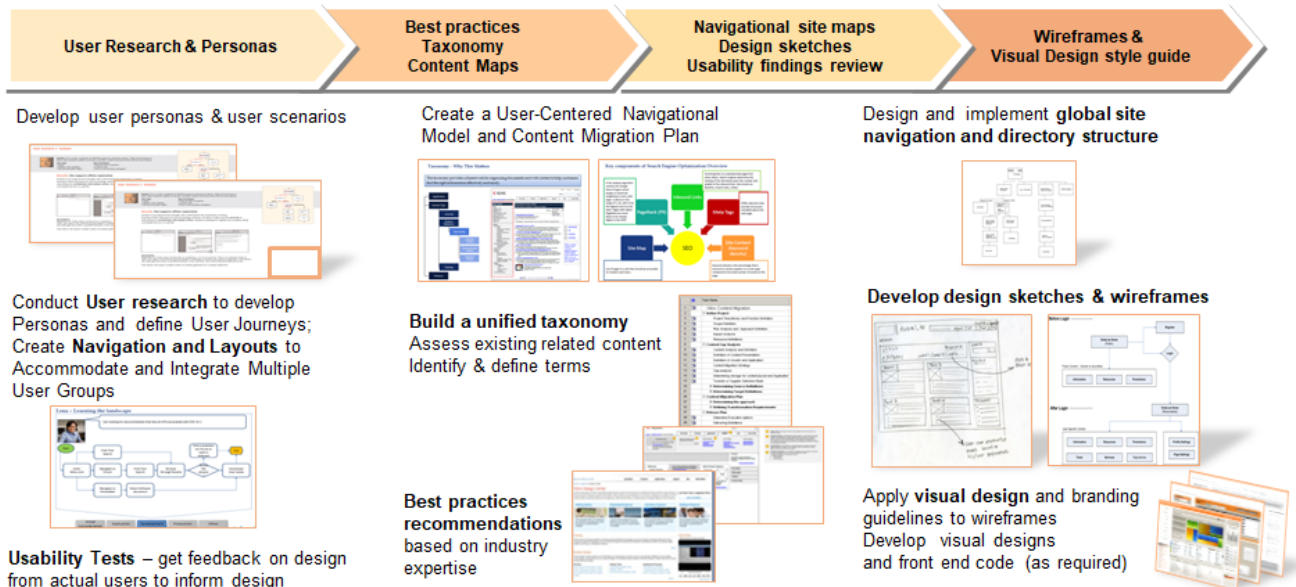


Figure 4 User interface design deliverable

We develop the user personas, taxonomy and content maps and best practices. We also develop the sitemap and usability findings. Finally we design the wireframe and visual design and style guides.

16.4.3 Implementation/Develop phase

The main activities during the implementation phase are as follows:

- Requirement gap analysis
- Gathering AS-IS security architecture, external applications
- Finalizing business rules
- Finalizing pseudo code and other documentation
- Finalizing integration points with other application
- Detailed system design
- Convert the design into complete system
- Versioning and source control
- Unit test case execution
- Assisting in Installing and setting up various environments
- Creating and testing databases as per scope
- Preparing unit test case procedures
- Coding & Compiling
- Performing test readiness review
- SSO integration & web- services.
- Content creation (static contents, templates, workflows)

The main deliverable from the implementation phase are as follows:

- Detailed System Design Document
- Configuration Mgmt. Plan
- Detailed Project Plan
- Configuration, customization coding and deployment
- Completion of project documentation
- Creation of manual document

- Deployment Guide

16.4.4 Testing phase

During the test phase we validate the functionality. We test the mobile app across all supported devices of various screen sizes and mobile OS platforms. Given below are the main activities of mobile testing:

- Requirement gap Analysis
- Business Scenarios gathering
- Create a Test Plan
- Creating Functional Test Cases
- Requirement Traceability Matrix
- Test Data Identification
- Test Execution
- Defect Management
- Status Report

16.4.5 Deployment phase

During this phase we deploy the mobile app to the mobile device and test the functionality. We test the mobile app functionality based on various user inputs. We also test the security and performance of the application on the actual device.

We test the screen usability across various screen sizes. Also we test the error handling capability of the app.

16.5 DEVELOPMENT TOOLS AND EMULATORS

Various integrated development platforms and emulators can be used for mobile app development.

Integrated development platforms provide various tools and services to boost the developer productivity for mobile app development. Integrated development environment provides all the features required for a developer in a single place. given below are the main features of a typical integrated development platform:

1. Source code editor: The visual editor of an IDE provides various features for developers searches autocomplete, Syntax highlighting code refactoring code snippets etc.
2. Code Compilation: The IDE compile the source code into binaries
3. Plugins: The IDEs provide various plugins for support specific deployment types, framework development and others.
4. Code debugging: The IDE allows developers to debug the code through various features such as stepwise execution variable value inspection and so on
5. Package and build: The IDE Provides features for building the code and packaging the code into deployable artifacts.
6. Integrated source control: The IDEs provide seamless feature to check out and check in the code from the source control systems.

Besides these the IDEs provide various other features such as search in files, boiler plate code, auto generation of code and such.

Visual Studio is one of the most popular tools for developing dot net applications. Android studio is the preferred tool for developing Android mobile applications. IntelliJ IDEA, Cordova, Eclipse, NetBeans are other popular IDEs.

Emulators are virtual devices that let developers test their mobile apps on various mobile devices.

Android studio is one of the most popular integrated development environments for Android mobile app development. The Android studio is free and provides smart code editors code Bin systems code profiling systems visual code editors to boost them developer productivity Android studio also provides emulators for testing the Android based mobile apps on various devices Andy, BlueStacks, Manyo, Xamarin, Visual studio Emulator are other competitors for Android development.

XCode is an integrated development environment for developing Apple iOS apps. AI can use XCode to develop apps for Apple Mac, Apple watch Apple iPad and Apple TV. Hex code provides an intelligent source code editor and the visual builder. the tool also provides code compiler code debugger and simulator.

☞ Check Your Progress 1

1. A ____ is a software program that runs on the CPU and uses the memory for computation
2. ____ is a lightweight process as it needs minimal resources
3. The mobile OS exposes the _____ for each of the system functions.
4. The five phases in the development process are _____
5. The solution design happens in ____ phase.
6. _____ are virtual devices that let developers test their mobile apps on various mobile devices.

16.6 APPLE iOS

Apple iOS 15 is the latest version of the I was that runs on various devices like Mac iPhone and iPad. Apple devices are immensely popular for the sleek design, highly intuitive user interface and for its performance. The key features iOS 15 or given below:

- Face ID recognition that unlocks the iPhone using user's face.
- FaceTime that allows users to connect and share with their friends and family and watch movies on their iPhone devices. The SharePlay allows users to share the movies and music during the FaceTime chat with their friends and family.
- Enhanced messaging where users can share the messages with their friends and family members.
- Customizable Memoji using colors glasses, colored headwear and other features.

- Enhanced focus and notification feature. The feature provides the users with focus time through do not disturb mode while doing activities such as reading, driving, fitness, sleep etc.
- Enhanced maps that provide interactive view of City for easier navigation
- Enhanced Safari browsing experience through tab groups, customizable start page, tab group synching, refresh by pull, voice search and others.
- Enhanced speech processing through Siri
- Enhanced accessibility features at libraries for Augmented reality, camera , gaming , keyboards , music notes, reminders, fonts, translate widgets and others.
- Enhanced gaming features such as multi-player games, game highlights,
- Enhanced camera features through improved Panorama pictures.
- Other intuitive features such as Live Text, Wallet, Spotlight, Photos, Health, Mail, Privacy, Apple ID, accessibility, App library, App store, CarPlay, dictionaries, keyboard dictation, translation, voice memos and others.

Apple iOS uses Swift, Cocoa and Cocoa touch for developing user interfaces. Apple iOS runs on Apple's ARM based chip. Safari is the default web browser on Apple IOS. Apple Safari supports multiple advanced features such as search, bookmarks, web suggestions, HTML5 autofill forms, OCR, web and native integrations. Apple iOS supports various network connectivity such as GSM, EDGE, Bluetooth, Wi-Fi, 3G, 4G and others.

Apple devices are known for highly intuitive and user-friendly interfaces. iOS supports personalized home screen, multi touch features, virtual keyboard, configurable widgets autocompletable, live wallpapers and others. IOS device drivers provide interfaces to various devices such as Global positioning system, accelerometer, magnetometer, camera and others.

Apple iOS also supports multitasking features. Users can switch across various iOS apps and work with them simultaneously. Apple uses iCloud for storing the user photos and media.

Developers can use IOS SDK for various activities such as database processing, XML parsing and so on.

16.6.1 IoS Architecture

The Apple iOS architecture consist of various layers as depicted in Figure 5. We primarily have user interface layer, media layer, core layer and kernel layer. Each layer has a distinct responsibility.

The user interface layer includes components that provide various user interface features such as touch, pinch, drag and others. The App Kit and touch framework handle the user inputs and gestures. We also have various apps like chat, email, FaceTime, Apple Pay and widgets like calendar, weather, stock quotes and others.

The media layer includes components to render images, video and other media files. We have libraries to render 2D graphics, 3D graphics, animation and others.

The core layer provides the foundation services, phone services, cloud services, sharing services and others. The kernel layer interfaces with the hardware and provides networking services. The networking services supports various network types such as Wi-Fi, NFC, CDMA, 4G etc. The device drivers provide interface to storage devices and USB ports.

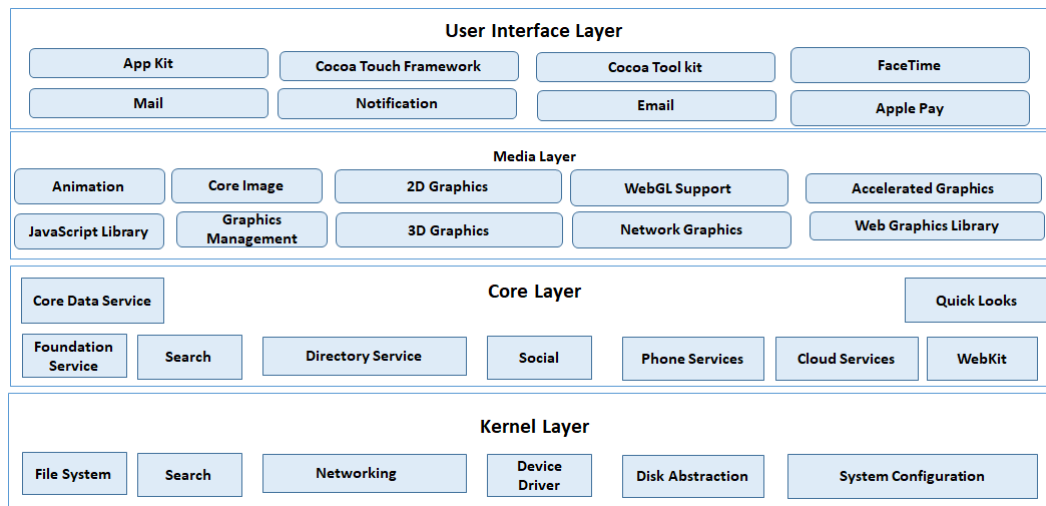


Figure 5 Apple iOS architecture

16.7 ANDROID

Android is one of the most popular mobile platforms and uses Linux. The Android code runs on Dalvik virtual machine Which is designed for mobile devices. Java programming language can be used for developing Android apps. java being an interpreted language its platform Independent and hence runs on the virtual machine. Android uses modified Linux kernel, licensed Apache server. Various device manufacturers such as Samsung, HP and others modify Android for their devices.

Android 12 is the latest release of Android. it provides a personalized experience for users so that they can use personalized widgets and personalize their home page. Android 12 provides refreshingly new design and responsive UI for the user interfaces. Android 12 supports accessibility features such as area magnification, bold text, grayscale and others. Android 12 he is also more secure and Protects user Private data. Android 12 also supports various enhancements that has camera access control, location security, privacy settings and others. it also offers again mode for richer gaming experience.

16.7.1 Android Architecture

We have depicted the Android architecture in Figure 6.

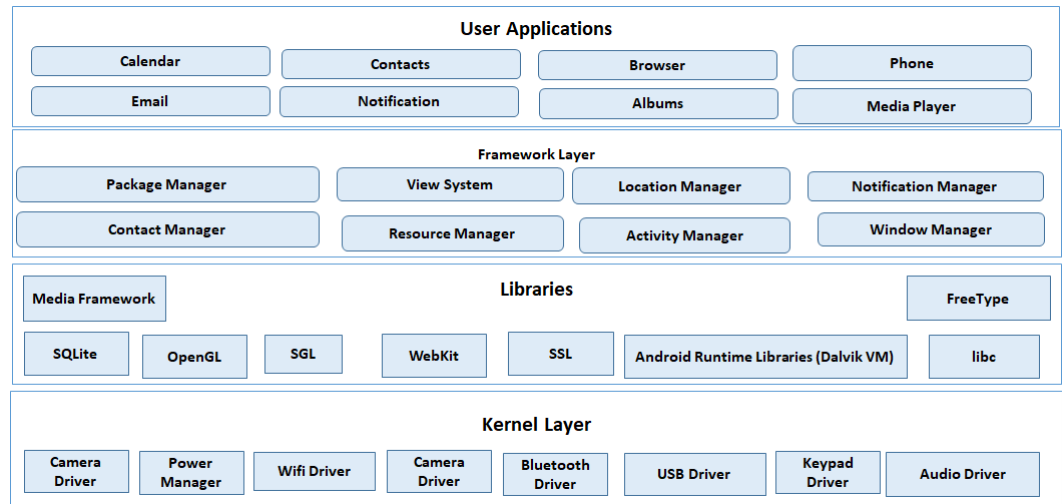


Figure 6 Android Architecture

Kernel Layer

Linux Kernel abstracts the hardware and provides all the device drivers for devices such as camera, keypad, USB, display device and others. Linux Kernel also handles the networking functionality. The kernel also handles the power and resource access.

Libraries

On top of Linux Kernel, we have various libraries such as the web browser webkit, libc (C Runtime library), OpenGL, FreeType for font rendering, SQLite database library, SSL library for handling the security, Surface manager for rendering windows and media framework for handling audio video and other media files.

Android also provides various input libraries for developers:

- android.app is a fundamental library provides access to application model
- android.content provides content management features such as content access, content publishing and messaging
- android.database provides database access features
- android.opengl provides 3D graphics rendering APIs
- android.text provides APIs for handling text
- android.webkit provide web browsing features

Android runtime includes Dalvik virtual machine which is optimized for Android. Dalvik VM uses Linux memory management and process management. Dalvik VM is optimized for mobile devices as it consumes less memory and has high performance.

Framework Layer

Application Framework layer provides high level API for the Android apps. Given below are the key services:

- Resource manager provide access to the color settings, user interface, layouts and other resources we use in the application.
- Notification manager helps in display alerts and notification
- Activity manager controls the application life cycle activity

- Content provider helps in publishing and sharing the data across applications
- View system helps in creating the user interfaces
- Telephony manager for managing the voice calls
- Location manager for managing the locations, GPS and cell towers

The layer also provides various APIs for locations, telephony, package management and others.

User Applications

We can develop number of Android apps such as ebook reader, games, SMS, calendar, browser using the application framework.

16.8 DIFFERENCES BETWEEN IOS AND ANDROID OPERATING SYSTEMS

Given below are the main differences between Android and iOS:

Category	Android	Apple iOS
OS Family	Linux	Unix, OSX
Source Model	Open Source	Commercial
Development Framework	Java	Swift
App Store	Google Play Store	App Store
Maps	Google Maps	Apple Maps
Voice Assistant	Google assistant	Siri
Data sharing and backup	Bluetooth	iTunes
Cloud Support	Google Drive	iCloud

Check Your Progress 2

1. _____ unlocks the iPhone using user's face.
2. _____ that allows users to connect and share with their friends and family and watch movies on their iPhone devices.
3. The _____ layer includes components to render images, video and other media file in Apple iOS
4. Android code runs on _____ virtual machine Which is designed for mobile devices.
5. _____ provides content management features such as content access, content publishing and messaging in Android

16.9 SUMMARY

In this unit, we started discussing the core concepts of the the OS. We looked at the key functions of mobile OS such as process management, memory management, device management and others. We also looked at the application programming interface exposed by the OS. We studies the five key phases of the development process including the requirements elaboration phase, design phase, implementation phase, testing phase and deployment phase. We studied various

IDEs and emulators such as Android Studio, Xamarin, Visual studio and others. We understood the key features of Apple iOS 15 such as Face ID recognition, FaceTime, enhanced camera and others. We also looked at various layers of the Apple iOS architecture such as user interface layer, media layer, core layer and Kernel layer. We looked at the main features of the Android 12 OS. We studied the Android architecture consisting of various layers such as user applications layer, framework layer, libraries and Kernel layer. We finally looked at the main differences between Android and Apple iOS

16.10 SOLUTIONS/ANSWERS

Check Your Progress 1

1. process
2. Thread
3. Application programming interface (API)
4. Requirements elaboration, design, develop and testing and deployment
5. Design
6. Emulators

Check Your Progress 2

1. Face ID
2. FaceTime
3. media
4. Dalvik
5. android.content

16.11 FURTHER READINGS

References

Mobile Computing 3rd Edition by Raj Kamal -
<https://www.amazon.in/Mobile-Computing-Raj-Kamal/dp/0199455414>